

### Exercice 1

$$a) N = \left\lceil \frac{\log_{10}((10^{AP/10} - 1) / (10^{AS/10} - 1))}{2 \log \frac{\omega_p}{\omega_s}} \right\rceil$$

$$= \left\lceil \frac{\log_{10}((10^{\frac{7}{10}} - 1) / (10^{\frac{16}{10}} - 1))}{2 \log(\frac{0,2\pi}{0,3\pi})} \right\rceil$$

$$= \lceil 2,78 \rceil = \boxed{3}$$

$$\omega_p = \frac{\omega_p}{(10^{AP/10} - 1)^{\frac{1}{2N}}} = \frac{0,2\pi}{(10^{\frac{7}{10}} - 1)^{\frac{1}{6}}} \approx 0,5$$

$$\omega_s = \frac{\omega_s}{(10^{AS/10} - 1)^{\frac{1}{2N}}} = \frac{0,3\pi}{(10^{\frac{16}{10}} - 1)^{\frac{1}{6}}} \approx 0,5$$

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0,5 + 0,5}{2} \approx \frac{1}{2} = 0,5$$

b) N est impair donc on a  $k = 0, 1, \dots, 5$  et  $\Theta = \frac{k\pi}{N}$

Il n'y a pas de zéros pour ce filtre

$$P_0 \quad \Theta = 0^0$$

$$P_1 \quad \Theta = \frac{\pi}{3}$$

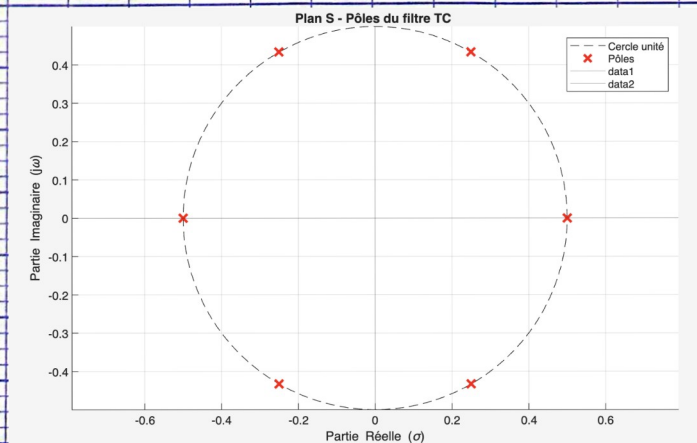
$$P_2 \quad \Theta = \frac{2\pi}{3}$$

$$P_3 \quad \Theta = \frac{3\pi}{3} = \pi$$

$$P_4 \quad \Theta = \frac{4\pi}{3}$$

$$P_5 \quad \Theta = \frac{5\pi}{3}$$

†



c)  $H(S') = \frac{\omega_c^3}{(S' - P_2)(S' - P_3)(S' - P_4)}$ ,  $P_2, P_3, P_4$  sont les pôles stables  
Calculons la valeur de ces pôles:

$$-P_2 = \omega_c e^{j2\pi/3} = \omega_c (\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3}) = -0,25 + 0,433j$$

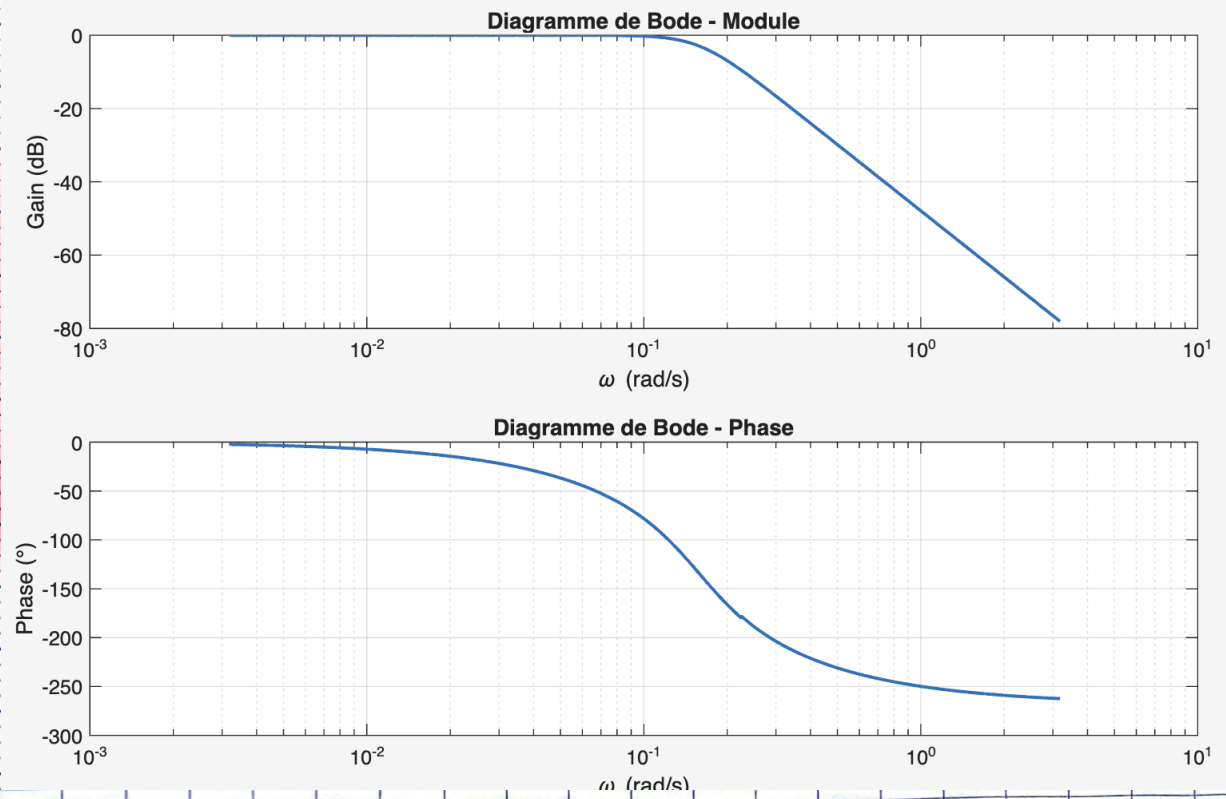
$$-P_3 = \omega_c e^{j\pi} = \omega_c (\cos \pi + j \sin \pi) = -0,5$$

$$-P_4 = \omega_c e^{j4\pi/3} = \omega_c (\cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3}) = -0,25 - 0,433j$$

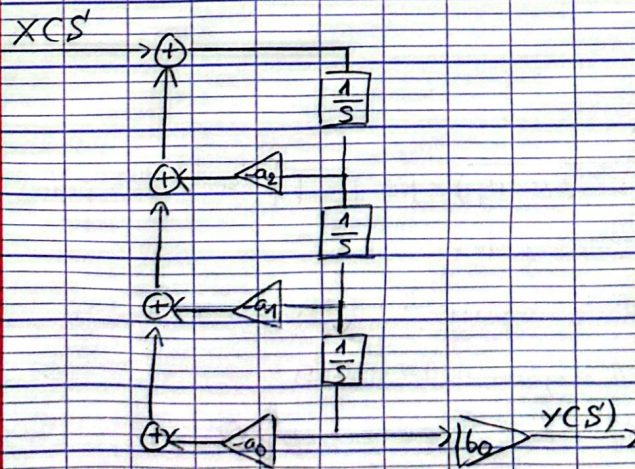
$$\text{Ainsi } H(S') = \frac{0,5^3}{(S' + 0,25 - 0,433j)(S' + 0,5)(S' + 0,25 + 0,433j)}$$



d)



$$\begin{aligned}
 e) H(s) &= \frac{(0,5)^3}{(s+0,25-j0,433)(s+0,5)(s+0,25+j0,433)} \\
 &= \frac{(b_0)/s^3}{(s^3 + a_2s^2 + a_1s + a_0)} \\
 &= \frac{b_0/s^3}{1 + \frac{a_2}{s} + \frac{a_1}{s^2} + \frac{a_0}{s^3}}
 \end{aligned}$$





## Exercice 2

$$a) \omega_p = 2 \tan \frac{\Omega_p}{2} = 2 \tan \frac{0,2\pi}{2} = 0,6498$$

$$\omega_s = 2 \tan \frac{\Omega_s}{2} = 2 \tan \frac{0,9\pi}{2} = 1,0190$$

$$N = \left\lceil \frac{\log_{10}((10^{\frac{1}{10}} - 1) / (10^{\frac{16}{10}} - 1))}{2 \log \left( \frac{0,6498}{1,0190} \right)} \right\rceil = \lceil 2,527 \rceil = 3$$

$$\omega_{cp} = \frac{0,6498}{(10^{\frac{1}{10}} - 1)^{\frac{1}{6}}} = 0,5154$$

$$\omega_{cs} = \frac{1,0190}{(10^{\frac{16}{10}} - 1)^{\frac{1}{6}}} = 0,5537$$

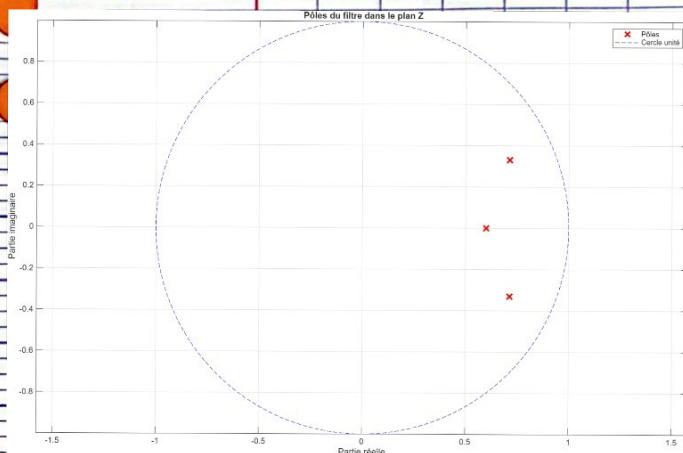
$$\omega_c = \frac{\omega_{cp} + \omega_{cs}}{2} = \frac{0,5154 + 0,5537}{2} = \frac{1,0691}{2} \approx 0,5$$

$$HCS) = \frac{(0,5)^3}{(s + 0,25 - 0,433j)(s + 0,5)(s + 0,25 + 0,433j)}$$

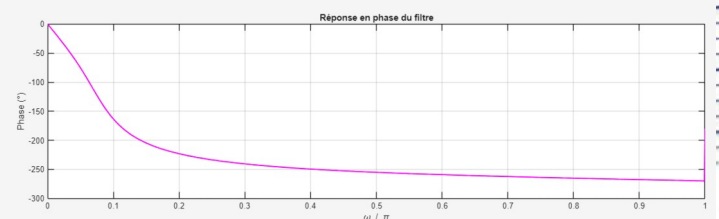
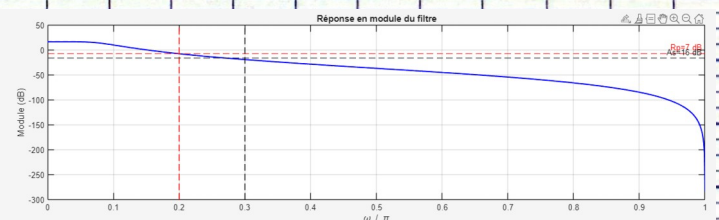
$$b) H[z] = HCS) \Big|_{s = 2 \left( \frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$= \frac{0,5^3}{\left( 2 \left( \frac{1-z^{-1}}{1+z^{-1}} \right) + 0,25 - 0,433j \right) \left( 2 \left( \frac{1-z^{-1}}{1+z^{-1}} \right) + 0,5 \right) \left( 2 \left( \frac{1-z^{-1}}{1+z^{-1}} \right) + 0,25 + 0,433j \right)}$$

$$= \frac{0,125}{13,125z^3 - 26,625z^2 + 19,375z - 4,875}$$

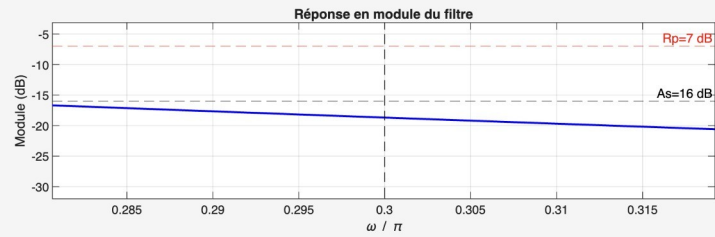
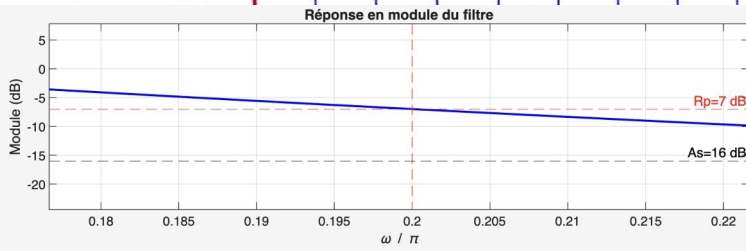


c)





## Vérifications des specs :



### Code :

```
Rs = 7;
As = 16;
omegap = 0.2 * pi;
omegas = 0.3 * pi;

wp = 2 * tan(omegap / 2);
ws = 2 * tan(omegas / 2);

N = ceil(log10((10^(As/10) - 1) / (10^(Rs/10) - 1)) / (2 * log10(ws/wp)));

wc = wp / ((10^(Rs/10) - 1)^(1/(2*N)));

poles = zeros(1, N);
for k = 1:N
    poles(k) = wc * exp(1j * pi * (2*k + N - 1) / (2*N));
end

den_a = real(poly(poles));
num_a = wc^N;
fs = 2;
[num, den] = bilinear(num_a, den_a, fs);

omega = linspace(0, pi, 1000);
Hz = zeros(size(omega));
for k = 1:length(omega)
    z = exp(1j * omega(k));
    Nz = 0;
    Dz = 0;
    for n = 0:length(num)-1
        Nz = Nz + num(n+1) * z^(-n);
    end
    for n = 0:length(den)-1
        Dz = Dz + den(n+1) * z^(-n);
    end
    Hz(k) = Nz / Dz;
end

% Ajuster le gain pour que le module à 0.2*pi soit -7 dB
[~, idx_07] = min(abs(omega - 0.2*pi));
gain_factor = 10^((-7 - 20*log10(abs(Hz(idx_07)))) / 20);
num = num * gain_factor;

% Recalcul après ajustement gain
for k = 1:length(omega)
    z = exp(1j * omega(k));
    Nz = 0;
    Dz = 0;
    for n = 0:length(num)-1
        Nz = Nz + num(n+1) * z^(-n);
    end
    for n = 0:length(den)-1
        Dz = Dz + den(n+1) * z^(-n);
    end
    Hz(k) = Nz / Dz;
end

mag_dB = 20 * log10(abs(Hz));
phase_deg = unwrap(angle(Hz)) * 180 / pi;

% Tracé
figure;
subplot(2,1,1);
plot(omega/pi, mag_dB, 'b', 'LineWidth', 1.5);
hold on;
grid on;
xlabel('\omega / \pi');
ylabel('Module (dB)');
title('Réponse en module du filtre');
plot([0.2 0.2], ylim, 'r--');
plot([0.3 0.3], ylim, 'k--');
yline(-7, 'r--', 'Rp=7 dB');
yline(-16, 'k--', 'As=16 dB');
hold off;

subplot(2,1,2);
plot(omega/pi, phase_deg, 'm', 'LineWidth', 1.5);
grid on;
xlabel('\omega / \pi');
ylabel('Phase (°)');
title('Réponse en phase du filtre');
```

